VIVEK BHARGAVA

Quantitative Researcher · Econometrics & Risk · Python & AI-Augmented Engineering
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EXECUTIVE SUMMARY

Work Experience	3.5+ years as Quantitative Researcher at the **Mathematical Finance Department, Fraunhofer Institute for Industrial Mathematics (ITWM)**, Kaiserslautern — production R and Python work for German insurance, Luxembourg banking, and German automotive clients. *Prior:* 2.5 years in Financial Advisory at PwC India and early-career engineering and operations roles.
Education	MSc Economics, University of Konstanz (Econometrics specialisation, grade 1.7, *Begabtenstipendium* 2021). MBA, Indian Institute of Management – Calcutta (Finance & Quantitative Methods). B.Tech, Indian Institute of Technology – Bombay (Mechanical Engineering).
Research Interests	Mixed-frequency models and reservoir computing; survival analysis for censored outcomes; multi-criteria portfolio optimisation; production deployment of statistical models. Agentic AI workflows for software engineering
Technical Stack	Production R (Shiny applications, package architecture, JDBC integrations against PostgreSQL/HANA, JVM-bridged numerical routines). Production Python (statsmodels and scikit-learn for econometric and ML modelling; PyTorch for research; pytest with full CI/CD). AI engineering: provider-portable LLM integration, structured-output prompt design, and a multi-surface agentic development pipeline (chat drafting → CLI implementation agent) with deterministic state transitions, human review gates, and CI-enforced quality gates.

CORE COMPETENCIES

Estimation & Inference — OLS, IV/2SLS, MLE, GMM, QMLE; Wald/LR/LM testing; bootstrap inference (parametric & asymptotic refinement); asymptotic theory (LLN, CLT, delta method); Monte Carlo simulation
Time Series & Structural Identification — VAR/VECM, SVAR (IV-identified, Blanchard–Quah long-run restrictions), cointegration (Johansen ML, ADF), Granger causality, impulse response & FEVD; Kalman filter & state-space; Bayesian VAR; MIDAS & mixed-frequency architectures
Volatility & Risk Econometrics — GARCH family (EGARCH, GJR, IGARCH, FIGARCH); VaR/CVaR/Expected Shortfall (parametric, historical, filtered historical simulation); backtesting (unconditional & conditional coverage); realised volatility; rolling/expanding-window evaluation
Statistical & Machine Learning — Regularised regression (Ridge/LASSO/Elastic Net), hierarchical & spectral clustering, Reservoir computing – Echo State Networks, Nested & time-series cross-validation, Feature selection algorithms, PCA, Softmax regression methods
Engineering — Production library design (30+ metrics, full test coverage); mixed-frequency time-series synchronisation; Pydantic v2 data modelling; repository-pattern & ports/adapters architecture; Streamlit & R Shiny operational dashboards; full CI/CD discipline (test, type, architecture-boundary linting)
Agentic AI Workflow Engineering — multi-surface AI development pipeline (chat ticket-drafting → CLI implementation agent) with deterministic state transitions and human review gates; per-module context engineering for token efficiency; production incidents distilled into reusable agent guardrails.

PROFESSIONAL EXPERIENCE

Fraunhofer Institute for Industrial Mathematics – ITWM *Nov 2022 – Present*
Quantitative Researcher, Mathematical Finance Department · Kaiserslautern, Germany

Multi-Criteria Portfolio Optimisation (German life insurer balance sheet)

- Owned the R Shiny decision-support layer of a production multi-criteria optimisation system for a German life insurer's asset allocation across 26 asset classes (return, volatility, regulatory capital, transaction costs) — radar plots, constraint sliders, Pareto scenario filtering. Built the feeding data pipeline, including regulatory-capital and HGB/IFRS accounting outputs.

Risk Engine Re-engineering — MATLAB → Python Migration (major Luxembourg bank)

- Re-engineered a major Luxembourg bank's legacy MATLAB risk engine into a production Python library now used by the bank's risk team — 30+ risk/return metrics (VaR, CVaR, Expected Shortfall, Sharpe/Sortino/Calmar) at full numerical parity with the legacy system, plus a mixed-frequency synchronisation layer handling missing data and irregular reporting periods. Shipped with full tests, GitLab CI/CD and API docs.

Budget Forecasting System — End-to-End Time Series Pipeline (insurance client)

- Built survival models on ~108K vehicle records (13K failures): a Weibull baseline extended with Random Survival Forests for nonlinear covariate effects without the proportional-hazards assumption. Documented the model's validity boundary (~1,300 days) so users would not extrapolate past where it was reliable.

Cashflow Modelling — Loan Disbursements & Repayments (insurance client)

- Co-developed a modular cashflow projection framework separating disbursement and repayment dynamics for forward-looking liquidity planning. Built empirical disbursement profiles by loan type and per-contract simulation of remaining outflows incorporating prepayment risk; designed a nonparametric repayment estimator handling left-truncation. **Joint paper in preparation:** "Nonparametric Loan Repayment Modelling Under Left-Truncation" (Fraunhofer ITWM).

Vehicle Failure Prediction — Survival Analysis & Ensemble ML (automotive OEM)

- Developed survival models on ~108K vehicle records (13K observed failures): Weibull parametric baseline for hazard estimation, extended with Random Survival Forests to capture nonlinear covariate effects without imposing the proportional hazards assumption. Documented model validity boundaries (RSF survival probability flattens beyond ~1,300 days) to prevent overconfident extrapolation

Prior Experience — India

- 2.5 years across financial advisory (PwC: cashflow models, scenario analysis and bid structuring for infrastructure PPP projects), technology operations (Housing.com: algorithmic call-routing), and petroleum logistics (Indian Oil Corp.). Additional internships at Reckitt Benckiser, Markem-Imaje and Tata Motors

PERSONAL PROJECTS

Research Stack — AI-Augmented Investment Research Platform

Solo-built Python platform · clean architecture · agentic AI workflow · 70+ tickets, 7 milestones, 867 tests

- **Agentic development workflow** — Designed and run a multi-surface AI pipeline: a chat agent drafts schema-constrained tickets; a CLI implementation agent executes them with deterministic state transitions, human review gates, and CI-enforced quality gates. Per-module context engineering for token efficiency; production incidents distilled into reusable agent guardrails.
- **Platform engineering** — Solo-built Python platform on a 4-layer clean architecture (ports & adapters), Pydantic v2 typed models, zero-I/O domain layer enforced by import-boundary linting; multi-source data layer (market-data APIs, FX feeds, broker CSV) via adapter/composite patterns with TTL caching.
- **Financial domain & scale** — German FIFO tax engine + pre-trade tax-impact simulator tested against hand-calculated references; FX-aware EUR-native valuation; correlation, concentration and performance analytics. Shipped as 70+ tickets across 7 milestones with an 867-test suite.

RESEARCH & ACADEMIC WORK

MSc Thesis — Reservoir Computing for Macroeconomic Forecasting with Mixed-Frequency Data

- Developed an Echo State Network (reservoir computing) architecture for macroeconomic forecasting using mixed-frequency data; benchmarked against MIDAS, UMIDAS, and Dynamic Factor Models. Proposed two novel state-equation modifications enabling mixed-frequency inputs; full from-scratch MATLAB implementation (reservoir dynamics, ridge-regularised readout, time-series cross-validation).

Selected Seminar Papers

- Causal Analysis of Monetary Policy Shocks: Structural VAR identified by external instruments
- Unsupervised Portfolio Choice by Clustering Financial Time-Series

EDUCATION

University of Konstanz *MSc Economics | Econometrics & Applied Economics · Grade 1.7 ·* *2019 – 2022*
Awarded Begabtenstipendium 2021 for academic excellence

Indian Institute of Management – Calcutta *MBA | Finance & Quantitative Methods* *2015 – 2017*

Indian Institute of Technology – Bombay *B.Tech Mechanical Engineering* *2009 – 2013*

LEADERSHIP & EXTRACURRICULAR (IIT BOMBAY)

- **Hostel Council Leadership**: Elected for 3 consecutive years; reached highest council position. Led a 2-tier team of 25 members across events, welfare, and hostel governance.
- **Theatre & Sports**: Acted in and directed 25+ theatre events; top positions at 10+ inter-college competitions. Inter-IIT Weightlifting rep. **Languages**: English (fluent, IELTS – 8/9) ·German (basic)